SOHAN GADEWAR

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Education

Vishwakarma Institute of Information Technology **2023 – 2027**
B.Tech in Information Technology | ***CGPA: 8.97/10.00*** *Pune, India*
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, System Design, Object Oriented Programming, Machine Learning.

Experience & Leadership

India Innovates Hackathon **Jan 2026 – Feb 2026**
National Finalist & Lead Builder — Ministry of Education & AICTE *Pune, India*
- Ranked among **top 20% finalist teams** nationwide out of **6,000+ participating teams** across India; spearheaded product discovery, scoping, and rapid cross-functional sprints.

GHCI 25 National GenAI Hackathon **2025**
Round 2 Qualifier — AnitaB.org India & Backbase ("Unbound with GenAI") *Credential Certificate*
- Selected for Round 2 in national GenAI challenge; engineered agent workflows and automated evaluation harnesses with structured JSON guardrails, validating prototype reliability under strict prompt constraints.

South African Patent Office (Filed Patent) **2025**
Patent Applicant & Inventor — AI-Powered Lost & Found Matching System *Status: Patent Filed*
- Conceptualized and filed patent for an AI matching architecture utilizing a **5-factor blind multimodal scoring engine** (image embeddings, NLP text semantics, spatio-temporal telemetry) with automated urgency detection.

Projects

Project Sentinel: AI Decision Memory & State-Drift Agent | *Python, PostgreSQL, FastAPI, Lovable, Notion* **Aug 2026 – Sept 2026**
Interactive Prototype: echo-sentinel-08.lovable.app | **PRD Specification:** Notion PRD Document
- Spearheaded product discovery across engineering standups, Slack, and Jira, identifying critical context loss from unprompted bot summaries.
- Formulated the **"Silent-by-Default"** product thesis, passively ingesting audio transcripts and commit streams without channel distraction.
- Architected deterministic audit engine powered by **append-only PostgreSQL schemas**, surfacing discrepancies between commitments and codebase state.
- Shipped clickable MVP on Lovable with live state-drift queries; authored comprehensive PRD detailing state transition matrices and query grammar.

CausalRetention: Prescriptive Uplift & Retention Decision Engine | *Python, XGBoost, Streamlit, Docker* **Oct 2026**
Live Decision Suite: Live Demo & Repository | **Technical Architecture:** CATE Uplift Spec
- Uncovered key economic friction in retention marketing where conventional churn models incur a **-13.1% net ROI** due to deadweight spend on non-responsive "Lost Causes".
- Architected prescriptive AI decision engine using a **Two-Model (T-Learner) XGBoost pipeline** to calculate CATE, segmenting customer cohorts into 4 distinct quadrants.
- Engineered a constrained **Knapsack financial budget optimizer** under a fixed $15K constraint, capturing **+$112.7K+ in net preserved revenue (+752% campaign ROI)**.
- Shipped containerized interactive Streamlit suite with Docker; validated incrementality via Qini curves and cumulative uplift against control baselines.

FlashDrop: High-Concurrency Flash Release & ACID Inventory Engine | *Java, Spring Boot, MySQL* **Oct 2026**
Source Code & Stress Suite: GitHub Repository & Benchmarks
- Architected a high-concurrency flash release inventory engine in Spring Boot backed by MySQL 8.0, eliminating race conditions and guaranteeing **zero overselling** via InnoDB pessimistic row-level locking (**SELECT ... FOR UPDATE**).
- Designed a **3NF relational schema with ACID stored procedures** (sp_reserve_product, sp_checkout_order), ensuring atomic multi-table state transitions across limited inventory, buyer holds, and finalized orders.
- Implemented defensive database triggers to prevent negative inventory anomalies and automated native **MySQL EVENT SCHEDULER** routines to reclaim expired 5-minute temporary cart reservations.
- Configured optimized **HikariCP connection pooling** with programmatic TransactionTemplate boundaries; verified via a multi-threaded Python stress suite simulating concurrent buyer races with **0% data inconsistencies**.

Technical Skills

Languages: Python, Java, SQL, Golang, C++, C, JavaScript
Frameworks: Spring Boot, FastAPI, Streamlit, XGBoost, Scikit-learn, React
Databases & Cloud: PostgreSQL, MySQL, Supabase, Docker, Git, Linux
Product & Tools: PRD Writing, Jobs-to-be-Done (JTBD), RICE, Product Metrics, Figma, Postman, Notion, LeetCode (400+ solved, 210+ GFG)

Certifications

- **Product Management:** Product Management Masterclass (GeeksforGeeks) — Product lifecycle, JTBD discovery frameworks, and roadmap metrics.
- **AI & Workflows:** Prompt Engineering | Advanced RAG (CampusX) | Docker for ML (CampusX) | n8n Automation (CampusX)